



## Lab 03

# Bipolar Junction Transistor (BJT) Characteristics

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### 3.1 Objective

To investigate the characteristics of Bipolar Junction Transistor and its input and output characteristic curves in Common Emitter configuration with load line

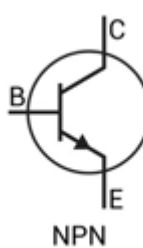
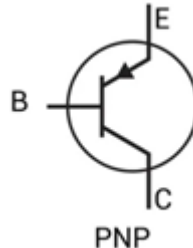
**Note:** All the plots required to be attached should be labeled properly, followed by Task number and must be readable, i.e., use white background, suitable marker color (recommended – any dark color) and increase the width of trace. On PSpice there is an option to add text labels in the plots.

### 3.2 Pre-Lab Task

#### Task 1: To draw the symbol of BJT and label its terminals and junctions

Name the three terminals and two junctions of BJT. Compare the terminals with respect to doping and thickness. Draw the symbols of *pn*p and *np*n BJTs with their terminals labeled and complete Table 3.1.

Table 3.1: BJT terminals and representation

| Terminals of BJT |           |                                                                                     | Doping                                                                                | Thickness        |
|------------------|-----------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------|
| 1                | Emitter   |                                                                                     | Lightly Doped                                                                         | Thin             |
| 2                | Collector |                                                                                     | Moderately Doped                                                                      | Most Thick       |
| 3                | Base      |                                                                                     | Heavily Doped                                                                         | Moderately Thick |
| Junction 1       |           |                                                                                     |                                                                                       |                  |
| Junction 2       |           |                                                                                     |                                                                                       |                  |
|                  |           | Symbolic Representation                                                             |                                                                                       |                  |
|                  |           |  |  |                  |
|                  |           | <i>npn</i> BJT                                                                      | <i>pn</i> p BJT                                                                       |                  |

### 3.3 Introduction to Bipolar Junction Transistor

Bipolar transistors are made of either silicon (Si) or germanium (Ge). Their structure consists either of two layers of n-type material separated by a layer of p-type material (*npn*), or of two layers of p-material separated by a layer of n-material (*pn*p). In either case, the center layer forms the base of

the transistor, while the external layers form the collector and the emitter of the transistor. The symbolic representation must have been obtained in Pre-Lab Task. Note that the arrow at the emitter terminal of the transistor symbol for either type of transistor points in the direction of conventional current flow and thus provides a useful reference. One part of this experiment will demonstrate how you can determine the type of transistor and its material, and identify its three terminals.

The relationships between the voltages and the currents associated with a bipolar junction transistor under various operating conditions determine its performance. These relationships are collectively known as the characteristics of the transistor which will be investigated in later part.

### Task 2: To identify the terminals, type and material of given BJTs

Figure shows a transistor structure encapsulated in plastic body. Such casing is usually called TO-92 package. The following procedure can be followed to identify the three terminals of a BJT along with its material (Si or Ge) and type (*nnp* or *pnp*).

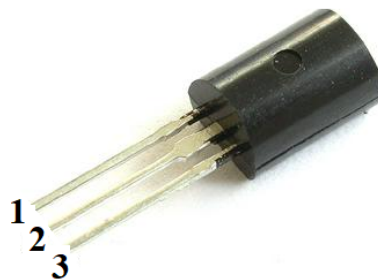


Figure 3.1: BJT in TO-92 package

### Procedure:

1. The transistor terminals are initially labeled 1, 2 and 3 with the position shown in figure. The multi-meter is operated in diode check mode.
2. The multi-meter probes are connected to each pair of terminals (1&2, 2&3 and 1&3) one by one in both polarities by reversing the probe connections.
3. For each of the above connections (listed in Table 2), the readings obtained using DMM in diode check mode are recorded.
4. If the given BJT is healthy, high (O.L. open loop or higher resistance) meter readings are obtained between two of the terminals regardless of the polarity of the meter leads connected. Neither of these two terminals will be the base as according to the structure, the base consists of different semi-conductor material (*p* type in *nnp* transistor and *n* type in *pnp*) from collector and emitter.
5. After identifying, base terminal, negative probe of DMM is connected to base and positive probe to either of the other terminals. If the obtained meter reading is low (approximately 0.7V for Si and 0.3V for Ge), the transistor type is *pnp*. If the obtained reading is high, the transistor type is *nnp*.
6. (a) For *pnp* type, negative probe is connected to the base terminal and positive probe alternatively to the each of the other two terminals. The lower of the two readings obtained indicates a connection between base and collector, the higher indicates connection between base and emitter.
7. (b) For *nnp* type, positive probe is connected to the base terminal and negative probe alternatively to the each of the other two terminals. The lower of the two readings obtained indicates a connection between base and collector, the higher indicates connection between base and emitter.



8. If the readings obtained in above steps are approximately 700 mV, the transistor material is silicon. If the readings are approximately 300 mV, the material is germanium. If the meter does not have a diode testing scale, the material cannot be determined directly. Apart from this, the same observations can be taken by operating DMM in resistance check mode as observed in Lab 01 with diodes as in forward bias, DMM registers low resistance as compared to that observed in reverse bias (O.L).

Perform the listed procedure on the given transistor (2N3904) and fill out the observations in Table 3.2.

Table 3.2: Data collection and results for identification of terminals, type and material for a given BJT

| Connection of DMM probes with BJT terminals        |          | DMM reading in diode check mode |
|----------------------------------------------------|----------|---------------------------------|
| Positive                                           | Negative |                                 |
| 1                                                  | 2        | 0                               |
| 2                                                  | 1        | 0.635                           |
| 1                                                  | 3        | 0                               |
| 3                                                  | 1        | 0                               |
| 2                                                  | 3        | 0.653                           |
| 3                                                  | 2        | 0                               |
| Identification of BJT terminals, type and material |          |                                 |
| Base Terminal                                      |          | 2                               |
| Transistor Type ( <i>pnp</i> or <i>nnp</i> )       |          | NPN                             |
| Collector Terminal                                 |          | 1                               |
| Emitter Terminal                                   |          | 3                               |
| Transistor Material (Ge or Si)                     |          | Si                              |

Based on the procedure listed above and for given identification of terminals in Table 3.3, predict the meter readings with a *pnp* transistor for all the listed connections.

Table 3.3: Prediction of data obtained in case of a *pnp* BJT

| Terminals, type and material of a given BJT |          |                                 |
|---------------------------------------------|----------|---------------------------------|
| Base Terminal                               |          | 2                               |
| Transistor Type                             |          | <i>pnp</i>                      |
| Collector Terminal                          |          | 3                               |
| Emitter Terminal                            |          | 1                               |
| Transistor Material                         |          | Ge                              |
| Connection of DMM probes with BJT terminals |          | DMM reading in diode check mode |
| Positive                                    | Negative |                                 |
| 1                                           | 2        | 0.29                            |
| 2                                           | 1        | 0                               |
| 1                                           | 3        | 0                               |
| 3                                           | 1        | 0                               |
| 2                                           | 3        | 0                               |
| 3                                           | 2        | 0.29                            |



### 3.4 Modes of Operation

There are 3 modes of operation of a BJT as given in Table which are categorized by the biasing of the two junctions; emitter-base (EB) junction and collector-base (CB) junction.

Table 3.4 a: Modes of operation of BJT

| Modes of Operation     | Base-Emitter Junction | Base-Collector Junction |
|------------------------|-----------------------|-------------------------|
| Cut-off Region         | Reverse-Biased        | Reverse-Biased          |
| Saturation Region      | Forward-Biased        | Forward-Biased          |
| Linear / Active Region | Forward-Biased        | Reverse-Biased          |

For *npn* configuration the conditions for base emitter junction and base-collector Junction Voltages in Reverse and Forward bias mode would be as follows:

Table 3.5b: Modes of operation of BJT

| Modes of Operation     | Base-Emitter Junction |             | Base-Collector Junction |             |
|------------------------|-----------------------|-------------|-------------------------|-------------|
| Cut-off Region         | Reverse-Biased        | $V_E > V_B$ | Reverse-Biased          | $V_C > V_B$ |
| Saturation Region      | Forward-Biased        | $V_B > V_E$ | Forward-Biased          | $V_B > V_C$ |
| Linear / Active Region | Forward-Biased        | $V_B > V_E$ | Reverse-Biased          | $V_C > V_B$ |

For *pnp* configuration the conditions for base-emitter junction Voltage and base-collector Junction Voltage in Reverse and Forward bias mode would be as follows:

Table 3.6 c: Modes of operation of BJT

| Modes of Operation     | Base-Emitter Junction |             | Base-Collector Junction |             |
|------------------------|-----------------------|-------------|-------------------------|-------------|
| Cut-off Region         | Reverse-Biased        | $V_B > V_E$ | Reverse-Biased          | $V_B > V_C$ |
| Saturation Region      | Forward-Biased        | $V_E > V_B$ | Forward-Biased          | $V_C > V_B$ |
| Linear / Active Region | Forward-Biased        | $V_E > V_B$ | Reverse-Biased          | $V_B > V_C$ |

To investigate the operation of BJT in these regions and relation between voltage and current at its input and output terminals, we will use the most frequently used transistor configuration called common-emitter configuration as the emitter is common to both the input and output terminals (in this case common to both the base and collector terminals) as shown in Figure 3.2. Two sets of

characteristics are necessary to describe fully the behavior of the common-emitter configuration: one for the input or base–emitter circuit and one for the output or collector–emitter circuit.

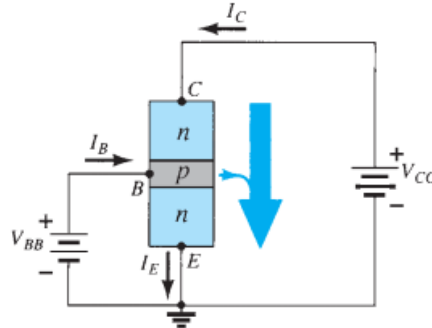


Figure 1: Common-emitter configuration

These characteristics are represented by obtaining VI curves i.e., base-current against base-emitter voltage ( $I_B$ - $V_{BE}$  curve) and collector current against collector-emitter voltage ( $I_C$ - $V_{CE}$  curve). When base-emitter junction gets forward biased, at a given base-emitter voltage a constant base-current is established. Output characteristics also called common-emitter characteristics i.e., plot of collector current against collector-emitter voltage, can represent the operation of BJT in different modes.

At small values of  $V_{CE}$ , the collector voltage is less than that of base causing CB junction to get forward biased. This causes the transistor to enter saturation region where both the junctions are forward biased. For a given base bias, increase in  $V_{CE}$  reduces the forward bias and eventually reverse bias the CB junction causing it to operate in active (linear) region. In this region, the collector current ideally should not increase once the CB junction becomes reverse biased. But practically due to early effect, the width of depletion region is affected by reverse biased voltage which increases the reverse saturation current. This makes the collector current to slightly increase at higher values of  $V_{CE}$  causing the characteristics to exhibit some slope.

### Task 3: To investigate the input characteristics of 2N3904 *n*pn-BJT using PSpice

Figure 3.3 shows 2N3904 *n*pn transistor connected in common emitter configuration. Its input characteristics can be investigated through plot between base current  $I_B$  and base-emitter voltage  $V_{BE}$  at a given collector-emitter voltage.

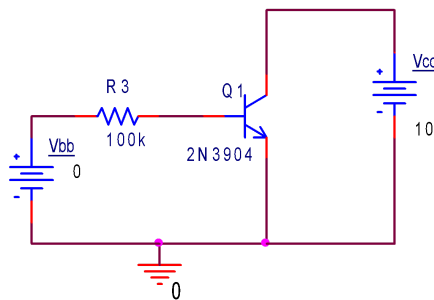
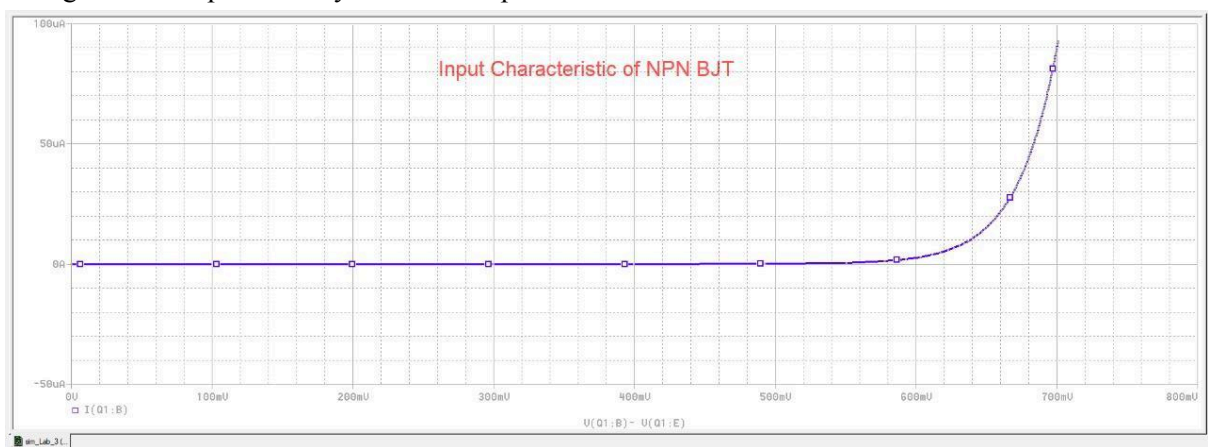


Figure 3.3: BJT 2N3904 connected in common-emitter configuration

1. Implement the circuit using OrCAD Capture. Set  $V_{cc}$  to 10 V which indicates that collector-emitter voltage is also set to 10 V as the source is directly connected to collector terminal.
2. To obtain curve between base current and base-emitter voltage, the base current will be varied by varying the voltage of source  $V_{bb}$ . To do this, perform DC sweep analysis taking  $V_{bb}$  as sweep variable. Vary its voltage from 0 to 10 V at an increment of 0.1V.
3. Run the simulation and obtain trace for base current. Set the x-axis variable to base-emitter voltage. Go to Axis Settings, X-Axis tab and change x-axis variable through the required expression e.g.  $V(Q1:B)-V(Q1:E)$ .
4. Attach the plot with suitable title and white background color. Analyze the input characteristics through obtained plots. Can you relate the plot with VI-characteristics of diode?

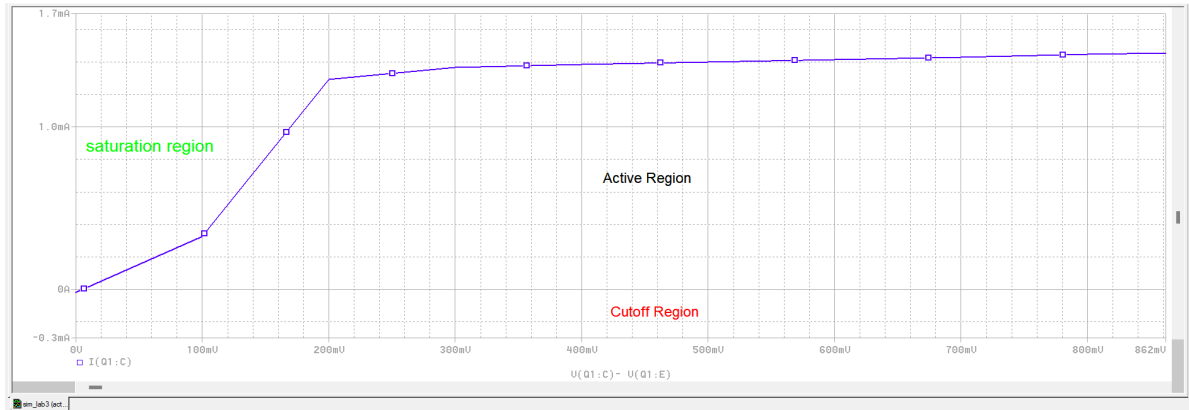


The base emitter junction of the transistor is a pn junction which behaves like a diode in forward bias therefore we see an exponential behaviour (beyond a certain cut-off voltage) in the trace obtained for  $V_{be}$  against  $I_b$  which resembles the VI characteristic of a diode.

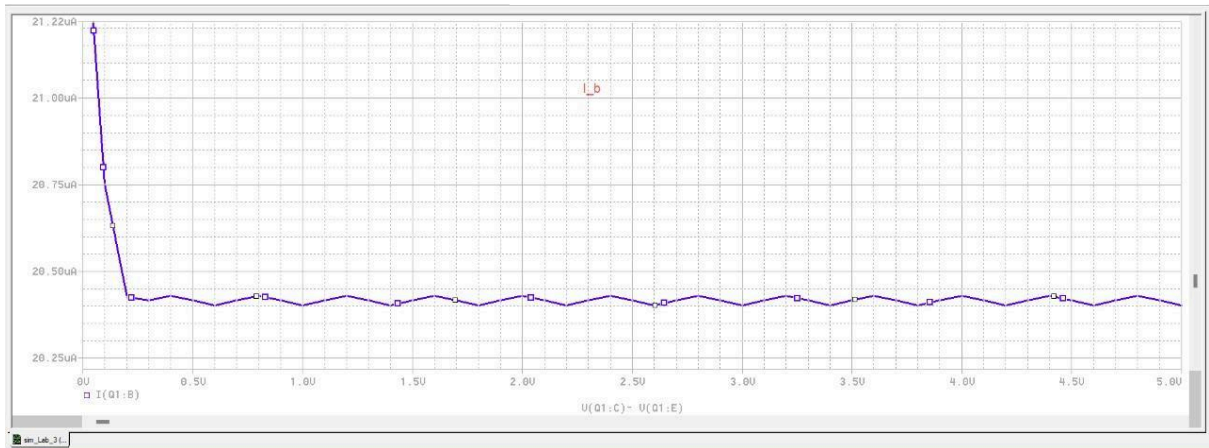
#### Task 4: To investigate the output characteristics of 2N3904 npn-BJT using PSpice

To investigate the output characteristics of BJT, collector current is plotted against collector-emitter voltage at different values of base current.

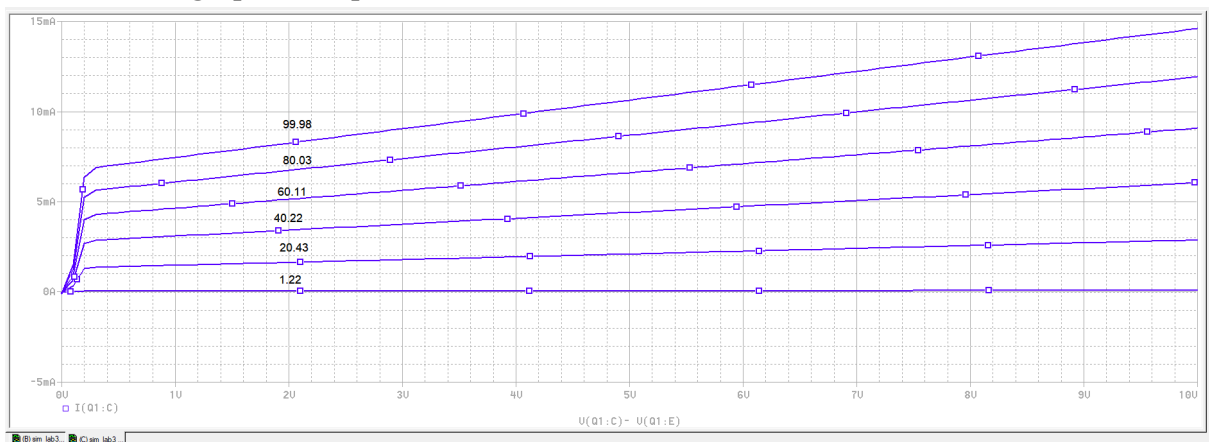
1. Modify the circuit implemented in previous task. Initially set  $V_{bb}$  to 2.7 V. Create a new simulation profile and perform DC sweep analysis to vary collector to emitter voltage by selecting  $V_{cc}$  as primary sweep variable. Vary this voltage from 0 to 10V at an increment of 0.1V. Run the simulation to obtain trace for collector current  $I_C$  and make sure the x-axis variable is set to  $V_{CE}$  (e.g.  $V(Q1:C)-V(Q1:E)$ ).
2. From  $I_C$ - $V_{CE}$  curve, identify the modes of operation of BJT and add suitable labels. **Attach** graph of the plot obtained **with labels** identifying the modes of operation and base current.



- Measure the base current also for which this plot is obtained. You can do this by adding trace for  $I_B$  in a separate window. **Attach graph of the plot obtained with labels.**



- Now, to investigate the effect of base current on this relation, the same primary sweep will be performed for different values of base current. To do this, edit simulation settings and check the secondary sweep option. Take  $V_{bb}$  as secondary sweep variable with its value ranging from 0.7V to 10.7V at an increment of 2V.
- Run the simulation and obtain  $I_C$ - $V_{CE}$  curves. With each curve **label** the corresponding base current. **Attach graph of the plot obtained with labels.**



- What is effect of increasing base current on the obtained collector current curves?





As the Base current increases, the collector current also increases. This is because the bipolar junction transistor operates under the principle that the collector current is proportional to the base current, following the relation

$$I_C = \beta I_B$$

$\beta$  is the current gain of the transistor. The transistor remains in active region as long as  $V_{CE}$  is sufficiently high. Increasing the base current shifts the  $I_C - V_{CE}$  curves upward, indicating a higher collector current for a given  $V_{CE}$ .

7. Analyze the relation between collector current with collector to emitter voltage in each mode of operation?

In the active region of the transistor, collector current increases with the emitter voltage due to the Early Effect of transistors. In the saturation region,  $I_C$  depends on  $I_B$  because the junction is in forward bias. In the cut-off region, the transistor is off therefore the collector current is negligible (approximately equal to 0V).

8. Does collector current remain constant in active region? What would have been the ideal behavior?

The collector current slightly increases with an increase in the emitter voltage due to the Early Effect but it does not remain constant in the active region. Ideally, the collector current would have been constant in the active region, independent of the emitter voltage since it is controlled by the base current.

### 3.5 DC Current Gain

In DC mode, the levels of  $I_C$  and  $I_B$  for any transistor are related by a quantity called beta and defined

$$\beta = \frac{I_C}{I_B}$$

by the following equation:

Where,  $I_C$  and  $I_B$  are determined at a particular operating point on the characteristics. For practical devices the level of  $\beta$  typically ranges from about 50 to over 400. This gain reveals the relative magnitude of collector current with respect to the base current. On specification sheet, it is usually included as  $h_{FE}$ .

#### Task 5: To analyze variation in DC current gains with collector-emitter voltage and base current

1. From the plots obtained in previous task, through cursors measure the collector current at two given points of each curve; when  $V_{CE}$  is 2V and when it is 10V. Record your measurements in Table 3.7 and mention corresponding base current.

Table 3.7: Data collection for DC current gain and its calculated values

| Base Current: $I_B$<br>( $\mu A$ ) | Collector-Emitter Voltage:<br>$V_{CE}=2V$ | Collector-Emitter Voltage:<br>$V_{CE}=10V$ |
|------------------------------------|-------------------------------------------|--------------------------------------------|
|------------------------------------|-------------------------------------------|--------------------------------------------|





|       | $I_C$ (mA) | $\beta$ | $I_C$ (mA) | $\beta$ |
|-------|------------|---------|------------|---------|
| 99.99 | 8.25       | 0.082   | 14.50      | 0.145   |
| 80.03 | 6.74       | 0.084   | 11.91      | 0.148   |
| 60.11 | 5.13       | 0.085   | 9.08       | 0.151   |
| 40.22 | 3.42       | 0.085   | 6.05       | 0.150   |
| 20.43 | 1.63       | 0.079   | 2.87       | 0.140   |

2. Calculate the DC gain  $\beta$  for each. Is there any variation in DC current gain as we move from left to right on the curve?

Yes,  $\beta$  varies slightly as we move from left to right on the curve i.e., as  $V_{CE}$  increases. This variation is due to early effect, which causes  $I_C$  to increase slightly with  $V_{CE}$ , even if  $I_B$  remains constant. Since  $\beta$  is the ratio of  $I_C$  to  $I_B$ , this results in a small increase in  $\beta$  at higher  $V_{CE}$  values.

3. This current gain  $\beta$  is the property of transistor and does vary by temperature as well. Go through the datasheet of 2N3904 BJT (<https://www.onsemi.com/pub/Collateral/2N3903-D.PDF>) and observe how this parameter is specified in the datasheet. The AC current gain is specified as  $h_{fe}$  in the datasheet (with small letters in subscript) which is something different than the DC gain. The curve shown in Figure 15 of datasheet illustrates the effect of temperature on  $h_{FE}$ . Analyze it briefly.

The DC current gain  $h_{FE}$  of the 2N3904 transistor, defined as  $h_{FE}=I_C/I_B$ , varies with collector current, collector-emitter voltage, and temperature. As shown in Figure 15,  $h_{FE}$  increases with temperature, as higher temperatures enhance carrier injection. However, at very low or high  $I_C$ ,  $h_{FE}$  declines. The Early effect causes a slight increase in  $h_{FE}$  at higher  $V_{CE}$ . Since  $h_{FE}$  is not constant, transistor circuits should be designed with temperature compensation and feedback techniques to ensure stability.

### Task 6: To investigate the change in operating point with the help of load line

The collector current together with collector to emitter voltage ( $I_C$ ,  $V_{CE}$ ) specifies operating point of transistor. The operating point is a fixed point on the characteristics, it is also called the quiescent point (Q -point). It is interesting to note that because the base current is controlled by the level of  $R_B$  and  $I_C$  is related to  $I_B$  by a constant  $\beta$ , the magnitude of  $I_C$  is not a function of the resistance  $R_C$ . Changing  $R_C$  to any level will not affect the level of  $I_B$  or  $I_C$  as long as we remain in the active region of the device. However, as we shall see, the level of  $R_C$  will determine the magnitude of  $V_{CE}$ , which is an important parameter.

1. To verify this, connect a 1k  $\Omega$  resistor at collector terminal of transistor. Set  $V_{bb}$  of 2.7V and  $V_{CC}$  at 10V. Simulate the circuit and obtain collector-to-emitter voltage and collector current and note down in Table.

Table 3.8: Operating point of BJT with 1k Ohm load resistor

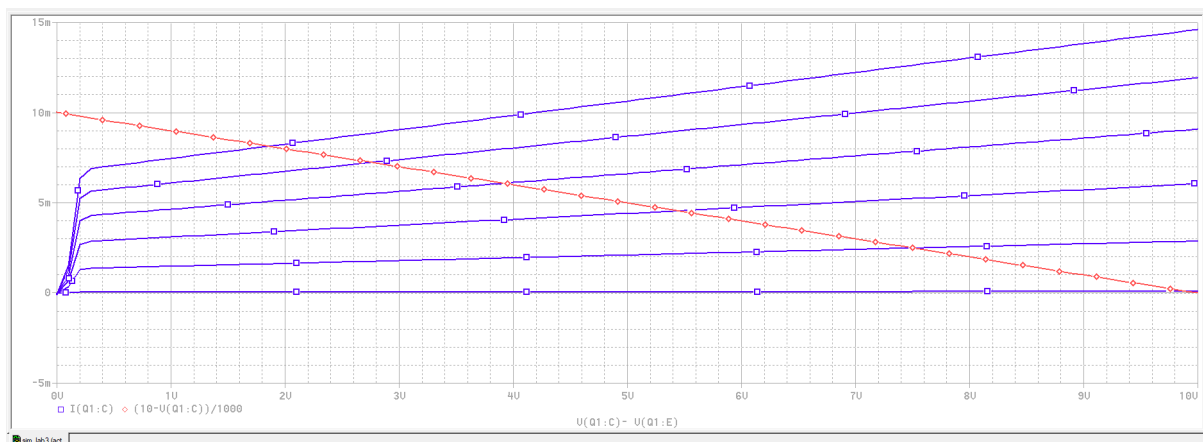
|                                     |          |
|-------------------------------------|----------|
| Collector-Emitter Voltage: $V_{CE}$ | 7.53 V   |
| Collector Current: $I_C$            | 2.490 mA |

- To directly obtain the operating point of transistor for a given load resistor and supply voltage  $V_{CC}$ , a load line can be plotted over the characteristic curves. The  $I_C$ - $V_{CE}$  characteristics curves obtained in Task 4 for different values of base current can be used to find operating point by adding load line. Remove the  $1k\Omega$  from circuit and through Primary and Secondary Sweep obtain  $I_C$ - $V_{CE}$  curves as done in Task 4 (or, re-use previously created simulation profile).
- To add a load line for  $1k\Omega$  load resistor  $R_C$ , add a trace with expression  $(10-V(Q1:C))/1000$ ,

$$I_C = \frac{(V_{CC} - V_{CE})}{R_C}$$

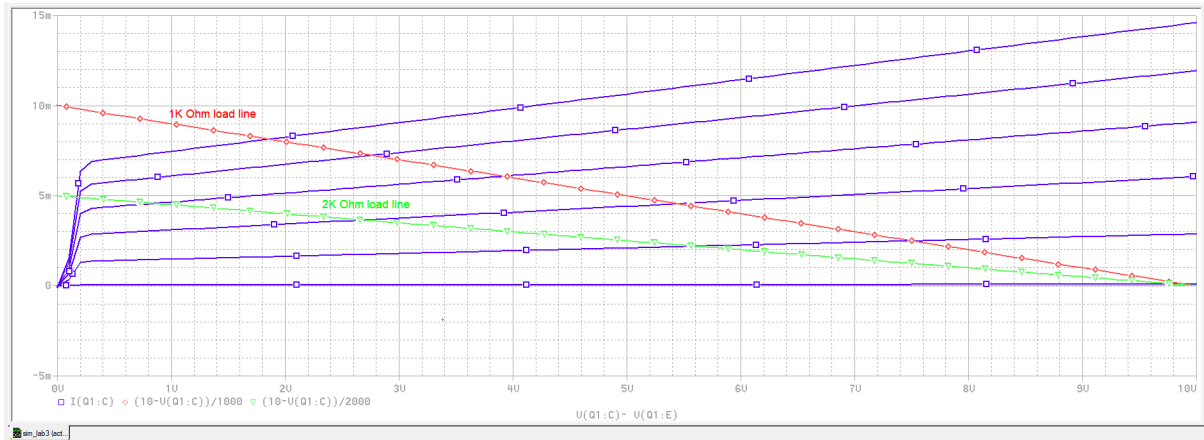
which is PSpice trace expression for  $R_C$  at fixed 10V  $V_{CC}$  and  $1k\Omega$  load resistor  $R_C$  when  $V_{CE}$  is being varied due to change in  $I_C$  as base current is varied. The intersection of the obtained load line with each characteristic curve gives the operating point of transistor for the corresponding base current. Note that instead of  $V_{CE}$ , collector voltage is used in PSpice trace expression as emitter is directly grounded.

- You can verify that the operating point found in Table 3.8 is the intersection point of load line and the  $I_C$ - $V_{CE}$  curve at base current of  $20\mu A$ . Attach the obtained plot and state your observations in the box below:



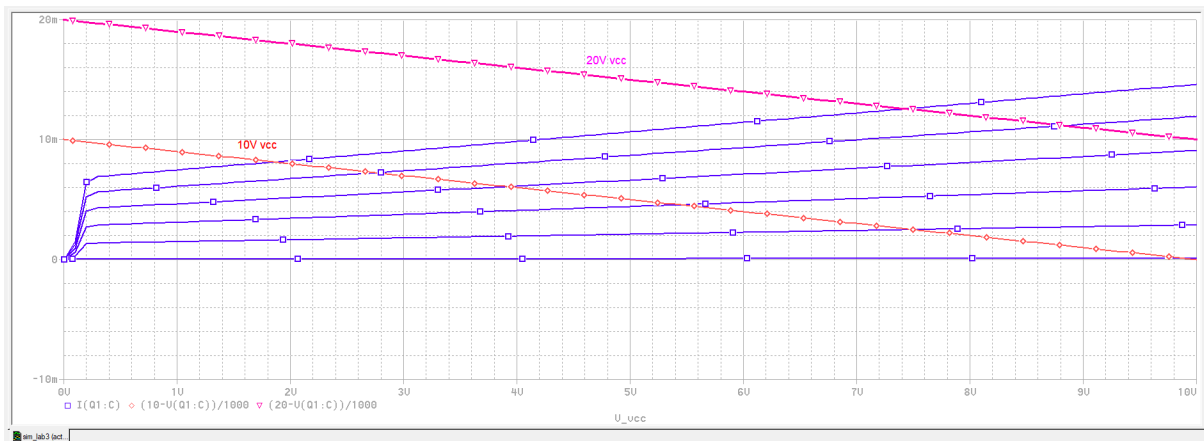
By plotting the load line on the  $I_C$ - $V_{CE}$  characteristics obtained in Task 4, the operating point (Q-point) for a  $1k\Omega$  load resistor and 10V supply is found at the intersection of the load line with the transistor curves. As  $I_B$  increases, the Q-point shifts higher along the load line, increasing  $I_C$  and slightly shifting  $V_{CE}$  lower. The transistor remains in the active region for moderate  $I_B$ , but moves into saturation if  $V_{CE}$  gets too low.

- To find the effect of varying load resistor, add another trace of load line with  $R_C$  of  $2k\Omega$ . Attach the capture of obtained plots with two load lines; one with  $1k\Omega$  and other with  $2k\Omega$  resistor. State your observations in the box below:



We can add the second load line using the following expression  $(10 - V(Q1:C)) / 2000$ . We can observe from the two load lines that With  $R_C=1k\Omega$ , the load line is steeper, allowing higher  $I_C$  and resulting in a lower  $V_{CE}$  for a given base current. In contrast, with  $R_C=2k\Omega$ , the load line is less steep, reducing  $I_C$  and increasing  $V_{CE}$ , shifting the Q-point upward on the characteristics. A smaller  $R_C$  moves the transistor closer to saturation, while a larger  $R_C$  keeps it in the active region longer, affecting gain and switching behavior.

- Now, to find the impact of varying supply  $V_{CC}$ , remove the load line for  $2k\Omega$  and add another load line with trace expression having  $20V V_{CC}$ . Attach the capture of plot.



- Analyze the impact of base current  $I_B$ , supply voltage  $V_{CC}$  and load resistor  $R_C$  on the operating point of transistor.

**Base Current ( $I_B$ ):** Increasing  $I_B$  shifts the Q-point upward, increasing  $I_C$  while slightly decreasing  $V_{CE}$ . If  $I_B$  is too high, the transistor moves into saturation.

**Supply Voltage ( $V_{CC}$ ):** A higher  $V_{CC}$  shifts the load line upward, increasing the available  $V_{CE}$  range. This extends the active region, allowing the transistor to handle higher voltage swings before reaching saturation or cutoff.



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**Load Resistor ( $R_C$ ):** A larger  $R_C$  makes the load line less steep, resulting in higher  $V_{CE}$  and lower  $I_C$  for a given  $I_B$ . Conversely, a smaller  $R_C$  steepens the load line, increasing  $I_C$  and pushing the transistor closer towards saturation.



**Assessment Rubric**  
**Lab 03**  
**Bipolar Junction Transistor (BJT) Characteristics**

|              |                    |
|--------------|--------------------|
| <b>Name:</b> | <b>Student ID:</b> |
|--------------|--------------------|

**Points Distribution**

| <b>Task No.</b>       | <b>LR2<br/>Simulation<br/>20</b> | <b>LR 4<br/>Data Collection<br/>22</b> | <b>LR5<br/>Plots<br/>24</b> | <b>LR 6<br/>Calculation<br/>4</b> | <b>LR 10<br/>Analysis<br/>24</b> |
|-----------------------|----------------------------------|----------------------------------------|-----------------------------|-----------------------------------|----------------------------------|
| <b>Task 1</b>         | -                                | -                                      | 4                           | -                                 | -                                |
| <b>Task 2</b>         | -                                | 12                                     | -                           | -                                 | -                                |
| <b>Task 3</b>         | 4                                | -                                      | 4                           | -                                 | 4                                |
| <b>Task 4</b>         | 8                                | -                                      | 8                           | -                                 | 8                                |
| <b>Task 5</b>         | -                                | 8                                      | -                           | 4                                 | 4                                |
| <b>Task 6</b>         | 8                                | 2                                      | 8                           | -                                 | 8                                |
| <b>CLO<br/>Mapped</b> | CLO 3                            | CLO 1                                  | CLO 3                       | CLO 1                             | CLO 1                            |

| <b>Affective Domain Rubric</b> |                    | <b>Points</b> | <b>CLO<br/>Mappe<br/>d</b> |
|--------------------------------|--------------------|---------------|----------------------------|
| AR 3                           | Pre-Lab Submission | 2             | CLO 4                      |
| AR 4                           | Report Submission  | 4             |                            |

| <b>CLO</b>   | <b>Total Points</b> | <b>Points Obtained</b> |
|--------------|---------------------|------------------------|
| 1            | 50                  |                        |
| 3            | 44                  |                        |
| 4            | 6                   |                        |
| <b>Total</b> | <b>100</b>          |                        |

*For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.*